

Kottapalli Bharath

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CAREER OBJECTIVE

Aspiring Mechanical Design Engineer with hands-on experience in CAD/CAE, FEA, and thermal systems analysis, Mechanical Systems Design & Analysis, seeking to contribute to innovative aerospace and mechanical organizations.

SKILLS

● Hard Skills:

- DESIGN & SIMULATION TOOLS: SolidWorks, CATIA V5, CREO Parametric, Auto CAD, ANSYS workbench, ANSYS Fluent, Open rocket.
- ENGINEERING EXPERTISE: CAD/CAE, Thermal Analysis, FEM, CFD, Rapid Prototyping, Rocket Design, Product Design.

● Soft Skills:

- CORE STRENGTHS: Problem Solving, Analytical Thinking, GD&T, Technical Documentation, Communication, proficient in Microsoft office 365.

WORK EXPERIENCE

Intern - Project Trainee

Jun 2025 – Jul 2025

ISRO - TELEMETRY TRACKING & COMMAND NETWORK CENTRE (ISTRAC)

Bangalore, KA

- Engineered a 12-way radial power combiner for X-band applications, enhancing deep-space data transmission by 2% and cutting procurement expenses by 95% through system integration.
- Developed an advanced heat-sink structure and performed ANSYS thermo-fluid analysis, achieving 99% cooling efficiency, minimized impedance, and improved operational reliability.

Rocket Design Intern

Jan 2024 – Mar 2024

FEYNMAN AEROSPACE LTD.

Mumbai, MH

- Designed and simulated single- and multi-stage rockets using SolidWorks and Open Rocket, achieving an apogee of 2,278 m, velocity of 361 m/s, and acceleration of 341 m/s² through optimized aerodynamic and propulsion modeling.
- Integrated and configured reloadable solid motors (Aerotech G-54W, Cesaroni 320H565-14A, Loki J820-LW) to refine thrust-to-weight ratio and ensure stable, efficient multi-stage flight performance.

CAD Engineer Trainee

Aug 2022 - May 2023

ALTIMECH ENGINEERING SOLUTIONS PVT LTD

Hyderabad

Duration: Aug 2022 - May 2023 (10 months)

- Where I have got placed as Cad Designer Trainee.
- Placed in detailing team for castings, machining and assembly drawings detailing in Creo Parametric 7.0
- Grip on Geometrical Dimensions and Tolerances (GD&T)

EDUCATION

HINDUSTAN INSTITUTE OF TECHNOLOGY AND SCIENCE (HITS)

Chennai, TN

Bachelor of Engineering, Mechanical Engineering

Expected April 2026

- Cumulative GPA: 8.88/10.0;
- Relevant Coursework: Thermodynamics, Strength of materials, CAD CAE, Heat and Mass transfer, FEM etc.

CENTRAL INSTITUTE OF PLASTICS ENGINEERING & TECHNOLOGY (CIPET)

Secunderabad, TS

Post Diploma in Plastic Mold Design using CAD/CAM
2023

Aug 2021-Jan

- Cumulative GPA: 9.14 / 10.0;
- Relevant Coursework: Plastic Materials, Mold design, product design, Rapid prototyping, CNC machining.

GOVERNMENT POLYTECHNIC COLLEGE.

Masabtank, Hyderabad, TS

Diploma in Automobile Engineering
2021

Jun 2018 - Apr

- Cumulative GPA: 9.2 / 10.0;
- Relevant Coursework: Thermal Engineering, Automobile Electrical Systems, CAD, Basics of Automobile Engineering.

NEWGEN SCHOOL OF EXCELLENCE

Hyderabad, TS

Matriculation – Graduation with Distinction

Jun 2015 – Mar 2018

- Cumulative GPA: 8.8 / 10.

ACADEMIC PROJECTS

INVESTIGATION ON HOT CORROSION BEHAVIOUR OF 304HCU STAINLESS STEEL EXPOSED TO MOLTEN SALT ENVIRONMENT

April 2026

- Tested the 304HCU stainless steel material coated with Na₂SO₄ – 75%NaCl under 750°C and conducted a corrosion comparative study on both Welded and parent material.
- We found that weld 304HCU corrodes 2 times faster than the parent metal with a 48.25% higher oxidation percentage. This happens mainly due to the chromium protective layer depletion in the metal happens at 750°C temp, under the reaction with Sulphate solution.

DESIGN AND DEVELOPMENT OF PORTABLE CHARGING STATION USING SUSTAINABLE ENERGY

May 2025

- Developed a portable charging station which can harness electricity from both wind and solar energies.
- This device can charge mobile accessories and trekking accessories in remote areas, is capable of generating up-to 65 W of electricity from wind and 60 W of electricity from solar energy.

EFFECT OF MATERIALS ON THERMO-MECHANICAL CHARACTERISTICS OF AN AUTOMOTIVE BRAKE DISC

Dec 2024

- Created and analyzed the thermo-mechanical properties of a 4-wheeler automotive ventilated brake disc using 4-different materials, (i.e, Grey Cast Iron, Al6061-T6, Ti-6Al-4V, and C/SiC)
- Through numerical simulation we observed that the metal Al6061-T6 has 138% high heat dissipation capacity when compared with the existing cast iron, followed by carbon ceramics with 92% and titanium alloy with 70% when compared with cast iron material.

DESIGN OF OMNIDIRECTIONAL WHEEL BASED MOBILE ROBOT

May 2024

- Created and animated an omnidirectional wheel-based robot using SolidWorks 3D modelling software and functioned animation monitoring the robot motion in all 360° rotation.

DESIGN OPTIMIZATION OF AERODYNAMICS ANALYSIS ON CAR BODY

Dec 2023

- Modelled and analyzed the aerodynamic effect on a modelled Nissan-GTR sports car which has modelled using the SolidWorks 3D modelling software.
- A Fluid analysis study was carried out to determine the effects of lift and drag coefficients of the car body at various speeds.

DESIGN OF TWO PLATE PLASTIC INJECTION MOLD

Jan 2023

- Prepared a detailed prototype of a two- plate plastic injection mold utilizing Creo; enhanced design accuracy, enabling a 20% reduction in material waste and conserving resource costs during production.
- A plastic flow simulation was studied to analyze the flow of molten plastic material in the mold and monitored the cooling rate and filling time, using CREO EMX tool.

CERTIFICATIONS & ACHIEVEMENTS

- Academic Excellence Award for 2025-26, Academic year
- ISRO-ISTRAC Internship Certificate – 2025
- ICT Design Week 2025 – Participant, Sri Sairam Engineering College, Chennai
- PALS Tinkathon 2024 – 25 – Project workshop at IIT Madras
- Rocket Design Internship Certificate – 2024
- Academic Excellence Award for 2023 - 24 & 2024 - 25
- 1st Prize – Designography, MechArena (20th National level Technical Symposium, Osmania University) – 2023
- AutoCAD 2D & 3D Training (CISTOCADD)

PUBLICATIONS

- “Studies on Radial Ventilated Disc Brake Materials – Effect of Metal Matrix Composite Materials using FEA in Automotive Brake Disc”, Advances in Additive Manufacturing Technologies, Taylor & Francis, 2024.

LANGUAGES

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| ● Telugu (Fluent, Native), | ● English, (Fluent) |
| ● Hindi, (Professional) | ● Tamil (Beginner) |